# Field2VC: Automating the Generation of ASPICE-Compliant Verification Criteria from Field Failures

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***Abstract*—Automotive ASPICE SYS.2 requirements must be verifiable, yet manually crafting verification criteria (VCs) that cover field-relevant failure modes is laborious and systematically incomplete. We present Field2VC, which grounds VC generation in two complementary failure knowledge sources: developer-reported defects from open-source repositories (GitHub) and consumer-reported field complaints from the NHTSA ODI database—two crowd populations operating at different abstraction levels. Structured failure-triple extraction and semantic-embedding matching connect each system requirement to relevant failure context, which a large language model synthesises into targeted, testable VCs. We contribute (i) an industrial benchmark of 350 de-identified ASPICE SYS.2 requirements with expert-annotated golden VCs (307 of which entered expert evaluation), and (ii) a four-expert blind evaluation of 489 generated VCs. Failure-grounded generation raises expert acceptance from a 45.3 % no-context baseline to 65.5 %–67.2 % across both channels (odds ratio 2.3–2.5×, Holm-corrected $p < 0.001$, Cohen's $h = 0.41$–$0.45$, small-to-medium effect), with consistent gains across body-control, communication-stack, and diagnostics requirement categories.**

***Index Terms*—verification criteria, automotive requirements, ASPICE SYS.2, issue mining, LLM, failure knowledge, ISO 26262, NHTSA**

## I. Introduction

In automotive software engineering, the cost of repairing a defect grows exponentially across the development lifecycle. Boehm and Basili's empirical survey finds that fixing a defect after delivery can cost up to 100 times more than fixing it during the requirements phase [1] (Fig. 1). In addition, ASPICE SYS.2 strictly requires every system requirement to be *verifiable*: a *Verification Criterion* (VC) must coexist with the requirement text [2].

Writing thorough VCs, however, fails in a specific and repeatable way. A requirement specifies intended behaviour under nominal conditions; a complete VC must additionally probe boundary conditions and fault modes. Reviewers reason forward from the requirement text, so the criteria they write can only confirm behaviours the specification already anticipates—while field failures, almost by definition, occur where it did not. The blind spot is structural rather than a matter of effort: a VC that omits an unforeseen edge case passes internal review precisely because no reviewer holds the evidence that would expose it (Fig. 1).

Closing this gap requires failure evidence from outside the project. Defect trackers are a natural candidate, and recent work such as CrUISE-AC [3] mines them to enrich agile acceptance criteria. For ASPICE SYS.2, however, the evidence must match the abstraction level at which the requirement is verified—the *system* level, not the software unit. Developer-reported defects describe faults as engineers observe them: isolated to a module, reproducible, often with a root cause attached. Failures that emerge from interactions across subsystems, environments, and months of customer usage rarely surface in any repository, because no developer observes the deployed fleet. Regulator-collected complaints capture precisely this stratum. NHTSA's ODI database records faults as the vehicle exhibits them, at fleet scale, under operating conditions no test plan enumerates [4], [5]. The two corpora are therefore not interchangeable: they sample the failure space at different abstraction levels, and our evaluation confirms the split empirically—complaint-grounded VCs are accepted more often for body and chassis requirements, while defect-grounded VCs dominate for communication and diagnostics modules (§V-B). We use NHTSA as a public proxy for the warranty and field-return data an OEM holds internally; the pipeline itself is agnostic to where the vehicle-level channel comes from. Manual inspection of 200 randomly sampled requirement–evidence pairs confirms the reliability of this evidence-mapping stage, yielding a precision of **73.5 %** (§IV).

This motivates our central question: given a system requirement, can generation grounded in cross-level field-failure evidence produce verification criteria that practising engineers would accept into a SYS.2 baseline? We treat blind expert acceptance as the measurable proxy for usefulness, since no oracle for "complete" VC coverage exists.

This paper presents **Field2VC**, an automated VC generation mechanism grounded in real failure evidence. Field2VC extracts defect context from two complementary domain channels: GitHub, which captures micro-level software failures, and NHTSA, which captures macro-level, vehicle-level failures. It filters domain-specific noise out of the raw records and uses the retained context as a semantic constraint, so that a large language model (LLM) generates verifiable VCs that cover historical failure scenarios.

*a) Contributions.:*

- **An automated, evidence-grounded pipeline** that combines developer-reported defects (GitHub) and regulator-collected complaints (NHTSA) as complementary evi-

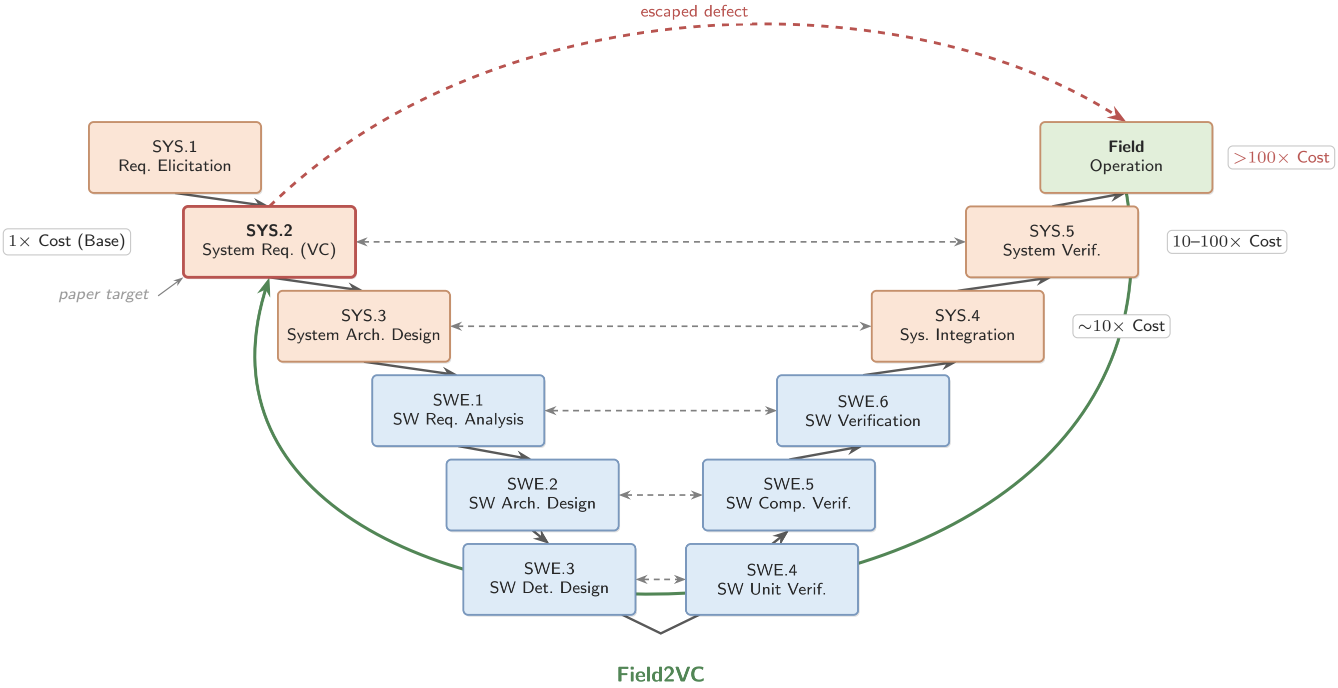


Fig. 1. ASPICE V-model cost escalation. A missing VC that passes SYS.2 review surfaces only at system-level testing or as a field complaint, where re-verification cost is 10–100× or more [1]. Field2VC shifts detection left by grounding VC generation in field-failure evidence *at* SYS.2 (highlighted).

dence channels at distinct abstraction levels, with an abstention mechanism that declines to generate when retrieved evidence is insufficient (§III).

- **A rigorous empirical evaluation.** Four independent experts evaluated 489 generated VCs spanning 307 unique industrial requirements, showing that VCs grounded in dual-channel evidence achieve significantly higher acceptance than a no-context baseline (§V).
- **A quantitative characterisation of the two evidence channels**, showing a systematic division of labour: the complaint channel performs better on body and chassis systems, the defect channel on communication and diagnostics modules (§V-B).
- **An industrial benchmark dataset.** We open-source a de-identified dataset of 307 real automotive system requirements—every item that entered expert evaluation—with their expert-annotated golden VCs, together with all generated VCs, per-rater labels, and analysis scripts, to support replication (Data Availability).

Our results extend work on issue-tracker-driven requirements enrichment [3] in two directions that prior work does not cover: the evidence is drawn from two channels at *distinct abstraction levels*, and the target artefact is an ASPICE-mandated verification criterion for a safety-relevant ECU rather than an agile acceptance criterion.

*b) Data Availability.:* To support double-blind review with a complete chain of evidence, the de-identified replication package is available *now* for review at https://anonymous.4open.science/r/apsec2026-artifact-C562/, containing: (1) the de-identified SYRS corpus covering all 307 evaluated items (drawn from a 350-item benchmark), with their expert-annotated golden VCs; (2) all 489 generated VCs with per-VC rater labels and majority-vote accept/reject labels, plus the adjudication record of the coordinating expert; (3) the evaluation guidelines (Framework B); (4) pipeline scripts for Stages 2–5 and all prompt templates; and (5) statistical analysis scripts. Proprietary signal names are masked by stable placeholders; the redaction script itself is part of the package.

## II. Related Work

### A. Automated Requirements Completion within Specifications

Requirements completeness research falls into two streams. *Refinement* approaches improve individual requirements through semantic enrichment without adding new external data, using techniques such as NL-level templating and domain-knowledge-driven pattern systems—for example, the EARS controlled requirements syntax [6]. *Derivation* approaches generate new requirements by exploiting relationships within or between existing requirement items.

LLM-based derivation has advanced rapidly [7]. ReqCompleter [8] fuses use-case models, E-R diagrams, and CRUD matrices into business-logic constraints that drive an iterative LLM completion process, achieving 20–88 % completeness improvement on seven real-world cases. F2SRD [9] retrieves the applicable OWASP ASVS verification requirements for each functional specification and guides GPT-4 to derive security requirements grounded in them. Both rely on *same-document* business logic as the completion signal; our work is complementary—we exploit *external* failure repositories to surface conditions that no single specification can reveal.

Closest in spirit is TVR [10], which applies retrieval-augmented generation (RAG) to validate and recover missing traceability links between stakeholder and system requirements in industrial automotive projects. TVR targets *traceability* (linking existing requirements) rather than *VC completion* (generating new complementary criteria); its retrieval corpus is the specification itself, whereas ours is external failure evidence. These two complementary directions indicate that RAG-based requirements engineering is an effective emerging paradigm across multiple automotive sub-tasks [11], [12].

### B. Enriching Requirements from External Repositories

Mining bug reports and issue-tracking systems for requirements engineering has a long history [13], [14]. Do et al. [15], [16] extract relevant requirements from similar software products using doc2vec and BIRCH clustering. Almonte et al. [17] use LLM-based prompting to indirectly derive quality-attribute non-functional requirements from functional specifications. In a similar vein, Koscinski et al. [18] use RelGAN to transform functional information into security requirements. These approaches mine requirements-like data from whatever related corpora are available; we instead mine two repositories from *different domains*—a defect-tracking database and a regulator-operated consumer complaint database—and show that the choice of knowledge source strongly predicts VC quality.

CrUISE-AC [3] is the leading crowd-data-driven system over external repositories. It mines public issue trackers (seven e-commerce systems, $>165$ k issue records, plus two content-management systems in a second domain) for issues matching Gherkin-format user-story acceptance criteria, achieving an 82 % approval rate in the e-commerce domain under a five-expert majority-vote evaluation with Gwet's $\mathrm{AC1} = 0.74$. Wang et al. [19] likewise generate acceptance criteria from multi-modal requirements data with LLMs, but target user-facing acceptance criteria rather than safety-critical system requirements.

Field2VC differs from CrUISE-AC in three significant respects. First, our domain is the automotive system-requirements level (ASPICE SYS.2) for safety-relevant ECU components. Second, we evaluate against a three-criterion Framework B standard (Completeness $\geq 3$, Novelty = Y, Usability = Y) rather than a binary acceptance judgement—a design that makes direct percentage comparisons misleading. Third, our ablation study contrasts LLM binary classification and embedding-based matching as grounding mechanisms for VC generation; it primarily exposes the effect of the filtering-enabled abstention decision on the results, rather than the matching method alone (§V-C).

### C. LLM-based Requirements Engineering

Recent work uses LLMs for diverse requirements tasks [20]–[23]: MARE [24] proposes a multi-agent collaboration framework for requirements elicitation; LATO [25] applies layerwise LLM reasoning to generate UML activity diagrams from requirements. Neither targets VC completion, and neither evaluates the effect of grounding in external failure evidence—the key contribution of Field2VC. In the automotive-safety domain specifically, LLMs have been applied to code generation and verification [26]–[28], functional-safety and security design [29], and RAG-based derivation of safety requirements from standards [30]; these operate on code or upstream safety requirements rather than completing verification criteria for existing SYS.2 specifications.

## III. FIELD2VC: TWO-CHANNEL VC COMPLETION PIPELINE

Field2VC operates as a five-stage pipeline that transforms raw failure records from two complementary external knowledge channels into candidate missing Verification Criteria (VCs) for automotive ECU SYRS items. Given a set of SYRS items and a target channel, the pipeline produces VCs grounded in empirical field-failure evidence, each traceable to one or more source records (Fig. 2). Throughout this section, we illustrate the pipeline using a real-world UDS Security Access requirement (SYRS.1018_B_B) and its corresponding field failure from GitHub (`python-udsoncan#51`), as detailed in the left column of Table I; the right column shows the analogous chain for Channel B, illustrating the channels' complementary roles—GitHub surfaces protocol-implementation boundaries, whereas NHTSA surfaces real-world interaction scenarios.

### A. Stage 1: Fetch

We define two channels providing complementary failure evidence:

**Channel A—GitHub Issues.** We index issues from four open-source repositories covering ECU-adjacent software stacks: `pylessard/python-udsoncan` (UDS diagnostics), `python-can-isotp` (ISO-TP transport), `hardbyte/python-can` (CAN bus), and `ecubus/EcuBus-Pro` (ECU test toolchain). We retrieved **696 issues** via the GitHub REST API,[1] prioritising closed issues labelled {bug, defect, regression, safety} and supplementing them with unlabelled closed issues for coverage. For the running example, this stage fetches `python-udsoncan#51`: *"SecurityAccess does not handle already-unlocked server."*

**Channel B—NHTSA Complaints.** We query the NHTSA ODI consumer-complaint database [31] using component keywords matching automotive ECU subsystems. We retrieved **364 complaint records**, each containing a free-text consumer narrative plus structured vehicle and component metadata. Unlike curated recall notices, complaint narratives are raw,

[1] https://docs.github.com/en/rest/issues

TABLE I

WORKED EXAMPLES (ONE ACCEPTED VC PER CHANNEL; SYRS, GOLDEN-VC, AND FAILURE TEXT QUOTED FROM THE DE-IDENTIFIED CORPUS AND PUBLIC SOURCES). GITHUB CONTRIBUTES A SPECIFICATION-SILENT PROTOCOL BOUNDARY; NHTSA A FIELD-FAILURE INTERACTION SCENARIO.

| | **Channel A — GitHub (SYRS.1018_B_B)** | **Channel B — NHTSA (SYRS.0190_B)** |
|---|---|---|
| SYRS item | "The system shall respond to the execution of RID `0x1008` within 50 ms. If the security preconditions (e.g., Security Access $27 L1 already unlocked) are not met, the system shall return NRC $33 (SecurityAccessDenied)." | "The system shall confirm a valid 'Light ON' condition if the sensor signal remains below the threshold for a continuous duration of $>$1000 ms. The headlight activation command shall be issued within 200 ms of this confirmation." |
| Existing golden VC | Tests the *locked* path only: ensure $27 L1 is LOCKED, send `$31 01 10 08`, verify NRC $33 within 50 ms (Negative Testing). | Tests the timing window only: reduce light below the threshold, confirm lights turn ON within $[1000, 1200]$ ms (Performance Testing). |
| Matched failure | `python-udsoncan#51` (sim. 0.69): per ISO 14229-1:2020 §10.4.1, an already-unlocked server answers requestSeed with an all-zero seed; the client instead derives a key from the zero seed and the server rejects it with NRC `0x24`. | ODI complaint #11557628 (sim. 0.63; 2022 Toyota Camry, EXTERIOR LIGHTING): "low beam headlights flashing when high beams are on … if you turn the car on with just ignition they will also flash." |
| Filtered triple | ⟨0x27 requestSeed while the level is already unlocked \| client should detect the all-zero seed and skip sendKey \| key derived from zero seed → NRC 0x24, protocol non-compliance⟩ | ⟨high-beam activation or ignition-on \| low beams remain steadily illuminated \| both low beams flash intermittently, state-stuck⟩ |
| Generated VC | *"Ensure $27 L1 is already UNLOCKED. Send $31 01 10 08. Verify the system does not return NRC $33 … and processes the RID 0x1008 within 50 ms"* (Negative Testing). | *"Simulate the activation of high beams while the light sensor signal is below the threshold … verify the activation command is issued within 200 ms of the 'Light ON' confirmation and the headlights do not flash"* (Fault Injection). |
| Expert verdict | **Accepted** ($C$=4, $N$=Y, $U$=Y; E1 adjudication) — a specification-silent boundary that both the SYRS and its golden VC omit. | **Accepted** ($C \geq 3$, $N$=Y, $U$=Y; unanimous) — an interaction scenario the timing-window golden VC never exercises. |

colloquial, and noisy field reports—which is precisely why the Stage-2 filter is architecturally necessary for this channel.

### B. Stage 2: Filter

Raw failure text varies in structure, length, and vocabulary. We use an LLM (Qwen-Plus) to filter each record into a structured triple:

⟨*Trigger* | *Protocol Behavior* | *Fault Mode*⟩

*Trigger* is the activating condition (e.g., a specific message sequence or timing event), *Protocol Behavior* is the expected ECU action per the relevant standard, and *Fault Mode* is the observed deviation. Filtering reduces vocabulary mismatch and suppresses implementation-layer details irrelevant to SYS.2 requirements.

The prompt's core instruction is *"Extract the technical essence from this [issue/complaint] report"*, with a mandatory escape route: if the record is not about automotive system behaviour (e.g., pure Python packaging, documentation), the model must return all three fields as `null`.[2] Records whose three fields return `null` are dropped here; this is where off-topic noise (packaging problems, documentation requests, pure consumer venting) leaves the pipeline. For the running example, issue #51 filters to the triple shown in Table I.

[2]All Stage-2 and Stage-4 prompts are released verbatim in the replication package (`prompts/`).

### C. Stage 3: Match

For each SYRS item $s$ and each filtered triple $t_i$ in the channel corpus, we compute cosine similarity [32]–[34] using the `text-embedding-v3` model (Alibaba DashScope; snapshot April–May 2026). We retrieve the top-$K = 5$ triples ranked by similarity, forming the *failure context* $\mathcal{F}(s) = \{t_1, \ldots, t_5\}$. We deliberately impose *no* hard similarity cutoff at this stage: every item proceeds with its top-5 context, and the relevance decision is delegated to the generation stage, where the model abstains (`NO_NOVEL_VC_FOUND`) if the retrieved evidence does not reveal a genuinely missing condition (§III-D). This mirrors, in post-retrieval form, the pre-retrieval relevance judgement that CrUISE-AC makes with an LLM binary classifier—a design choice our ablation study examines directly (RQ3, §V-C). For SYRS.1018_B_B, issue #51 ranks second at cosine 0.69 and enters the top-5 failure context.

Embedding-based matching provides two advantages over LLM binary classification (used in CrUISE-AC [3]): (1) it scales to thousands of (SYRS, issue) pairs without incurring per-pair LLM calls, and (2) it produces a continuous, auditable relevance score that is released with every match in our artifact.

### D. Stage 4: Generate

We prompt Qwen-Max with three inputs: the SYRS item, its existing golden VCs (as *do-not-repeat* context that operationalises "missing"), and the failure context $\mathcal{F}(s)$. The

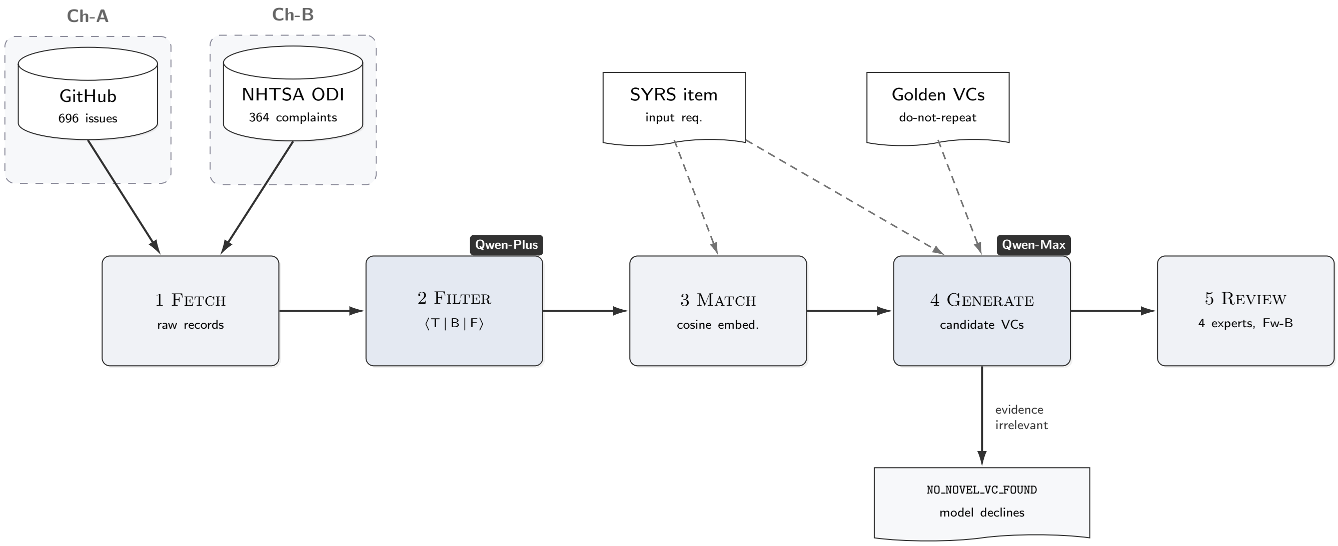


Fig. 2. Field2VC five-stage pipeline. Failure records from GitHub (Ch-A) and NHTSA (Ch-B) are filtered into ⟨Trigger | Protocol-Behavior | Fault-Mode⟩ triples (Stage 2, Qwen-Plus), embedding-matched to each SYRS item via cosine similarity (Stage 3, `text-embedding-v3`), and used to prompt VC generation (Stage 4, Qwen-Max). The generator is never forced to produce a VC: it may decline by emitting the fixed marker `NO_NOVEL_VC_FOUND` when the evidence reveals no genuinely missing condition—the pipeline's only relevance gate. Four domain experts review the surviving VCs under Framework B (Stage 5).

prompt instructs the model to generate only VCs that (a) genuinely test the requirement (not the protocol library the failure was reported against), (b) are not redundant with the existing golden VCs, and (c) are inspired by a real failure pattern in $\mathcal{F}(s)$ and testable at ECU system integration level. The prompt's core instruction is *"generate MISSING verification conditions … 'Missing' means test scenarios NOT already covered by the existing golden VCs"*, and it closes with an explicit permission to decline: *"If no genuinely useful missing VC can be identified from these issues, output: NO_NOVEL_VC_FOUND."* We refer to this fixed marker as the *sentinel*. Declining is an expected outcome, not an error: the model is never forced to produce a VC, so when the retrieved evidence is irrelevant the pipeline abstains instead of hallucinating a plausible-sounding criterion [35]. In the running example, the model turns issue #51's triple into the already-unlocked test of Table I—a path neither the SYRS nor its golden VC exercises.

### *E. Stage 5: Review*

Each generated VC is independently scored by at least two of the three external raters (E2, E3, E4) on three dimensions under **Framework B**:

- **C (Completeness)**: Score 1–5. Does the SYRS item omit this VC? ($C \geq 3$ is the acceptance threshold.)
- **N (Novelty)**: Y/N. Is this VC absent from the existing SYRS item and its associated VCs?
- **U (Usefulness)**: Y/N. Is this VC actionable for downstream test design and software qualification?

A VC is accepted if and only if $C \geq 3$ AND $N = \text{Y}$ AND $U = \text{Y}$. The label is the majority vote among the raters who scored each VC; E1 (adjudicator) resolves disagreements on the N and U dimensions. The running-example VC was accepted ($C$=4, $N$=Y, $U$=Y) after E1 adjudicated a novelty split between E2 and E3 (Table I).

### *F. Ablation Variants*

Two ablation conditions isolate the contribution of filtering and embedding-based matching:

**Abl-NF (No Filter):** Skip Stage 2. Embed raw issue title + body (first 500 characters) directly and match to SYRS via cosine similarity. This removes the structured triple extraction, testing whether raw-text embedding suffices.

**Abl-LM (LLM-Match):** Replace Stage 3 embedding similarity with an LLM binary relevance classifier (Qwen-Plus), replicating the CrUISE-AC matching approach [3]. This tests LLM judgment against embedding similarity as a pairing mechanism.

## IV. Experiment Design

### *A. Research Questions*

- **RQ1 (Effectiveness):** Does issue-driven prompting generate VCs with significantly higher expert acceptance than a no-context baseline?
- **RQ2 (Channel Analysis):** Which knowledge channel produces the highest-quality VCs, and how do they differ in acceptance rate, novelty, and usefulness?
- **RQ3 (Ablation):** What is the contribution of (a) failure-triple filtering and (b) embedding-based matching to VC quality?

TABLE II
DATASET STATISTICS

| Condition | VCs generated | Sources |
|---|---|---|
| Channel A (GitHub) | 87 | 696 issues |
| Channel B (NHTSA) | 128 | 364 complaints |
| Baseline (no-context) | 274 | — |
| **Total** | **489** | |

The 489 evaluated VC instances span 307 of the 350 SYRS items; for the remaining 43, the channel runs emitted only the abstention sentinel, so no VC entered evaluation. The two channels share no SYRS.

- **RQ4 (LLM Sensitivity):** How stable are the channel-level results across different LLMs from multiple model families?

## B. Dataset

*a) SYRS corpus.:* We apply the pipeline to 350 unique automotive ECU SYRS items drawn from a proprietary specification corpus of 675 items, covering body control (ESCL, EPB, SCCM), communication stack (UDS, CAN), diagnostics, and system state management components. The channel mapping is disjoint by construction: Ch-A covers 208 communication-stack and diagnostics items; Ch-B covers 142 body-control, HMI, and functional-safety items. All items follow the ASPICE SYS.2 format with explicit actor, trigger, quantitative boundary, and expected behaviour, and each carries human-annotated golden VCs.

*b) Generated VCs.:* The full pipeline generates 489 VCs across two channels and a baseline:

## C. Evaluation Metrics

*a) Accept rate.:* Fraction of VCs whose majority-vote label is Accept (Framework B: $C \geq 3$ AND $N$ = Y AND $U$ = Y).

*b) Wilson 95 % CI.:* Confidence interval for the accept rate using the Wilson score method [36].

*c) Fisher's exact test + Holm–Bonferroni correction.:* Each channel is tested against the baseline with Fisher's exact test [37]; $p$-values are corrected with Holm–Bonferroni [38] across three simultaneous comparisons (Ch-A, Ch-B, and A+B combined vs. baseline).

*d) Cohen's h.:* Effect size for proportion comparisons [39]: $h \geq 0.2$ small; $h \geq 0.5$ medium; $h \geq 0.8$ large.

*e) Generation-rate proxy.:* When the LLM determines that the retrieved failure context offers no test angle not already addressed by existing golden VCs, it is prompted to abstain—emitting the sentinel string `NO_NOVEL_VC_FOUND` instead of a VC. The *generation rate* is the fraction of SYRS items for which the model generates a VC (i.e., does *not* abstain); a high generation rate under active filtering indicates the model is finding genuinely new failure angles, while near-universal generation (>93 %) signals the model is ignoring the abstention instruction. We use the generation rate as a lightweight quality proxy for LLM sensitivity runs where full human evaluation is not available (RQ4).

*f) Abstention rates.:* In the primary evaluation, Ch-A (GitHub) abstains on 121 of 208 SYRS items (58.2 %); Ch-B (NHTSA) abstains on 13 of 142 items (9.2 %). The large difference reflects the vocabulary alignment gap: NHTSA complaint records, once filtered into structured fault modes, map tightly to the component-level language of body-control and chassis SYRS items, while GitHub issue vocabulary is more varied relative to communication-stack and diagnostics items, triggering abstention more often. This asymmetry is itself a qualitative finding: the pipeline selectively generates VCs rather than producing output indiscriminately.

*g) Matching precision.:* To establish confidence in the upstream retrieval quality, we manually evaluated a stratified random sample of 200 requirement–evidence pairs (100 per channel, seed = 42, similarity range 0.45–0.78). A pair is labelled *relevant* if the retrieved evidence describes a failure mode or boundary condition observable in the ECU under test (not merely a topically adjacent software defect or host-toolchain issue). The pipeline achieves an overall matching precision of **73.5 %** (Ch-A GitHub: **61.0 %**; Ch-B NHTSA: **86.0 %**), confirming that the generation step is grounded in failure-relevant semantics.

## D. Inter-Annotator Agreement

*a) Rater profiles.:* We recruited four raters to span the three stakeholder roles in ASPICE V&V: process ownership, academic methodology, and hands-on practice. All have direct automotive software engineering experience; their disciplinary mix—two industry practitioners, one process/safety specialist, and one academic researcher—guards against single-perspective bias in a multi-criterion framework. **E1** (Embedded Systems Project Manager, 10 yr industrial experience) acts as adjudicator, resolving split votes using defined criteria. **E2** (ASPICE assessor and functional-safety consultant) contributes direct SYS.2 process authority. **E3** (PhD candidate in automotive software testing) provides systematic, methodology-grounded judgement. **E4** (Senior functional-test engineer, automotive OEM/Tier-1) anchors evaluations in physical V&V executability. This industry–academia panel structure replicates the design of CrUISE-AC [3].

*b) Main evaluation (RQ1/RQ2).:* E2, E3, and E4 independently scored the 489 VCs such that every VC received at least two independent external ratings (266 VCs scored by two raters and 223 by three); E1 (adjudicator) resolved the cases of disagreement. The accept/reject label is the majority vote among the raters who scored each VC. We use Gwet's AC1 [40], [41] rather than Cohen's $\kappa$ to avoid the kappa paradox [42] that arises from the high prevalence of the Accept class under Framework B. Pairwise AC1 is reported on the binary full-accept criterion.

*c) Ablation evaluation (RQ3).:* The 50-item subset comprises the first 50 Ch-A items in corpus iteration order; all are Communication-Stack requirements. This homogeneity strengthens internal comparability across ablation conditions (category cannot confound the contrast) but restricts RQ3's

TABLE III
RQ1: ACCEPT RATES VS. BASELINE (THREE-CRITERION PROTOCOL; MAJORITY VOTE)

| Condition | VCs | Accept | Rate | Holm-$p$ |
|---|---|---|---|---|
| Baseline (no-context) | 274 | 124 | 45.3 % | — |
| Channel A (GitHub) | 87 | 57 | 65.5 % | 0.0007*** |
| Channel B (NHTSA) | 128 | 86 | 67.2 % | <0.001*** |
| **A+B combined** | 215 | 143 | 66.5 % | <0.001*** |

external scope to that category (§VI). All four raters (E1–E4) independently score the 50-item Ch-A ablation subset; the accept/reject label is the majority vote of E2, E3, and E4 ($\geq$2/3) with E1 as adjudicator (matching the main-evaluation protocol), and all four raters' scores are used for the pairwise AC1 analysis.

### E. Baselines

*a) No-context baseline.:* Qwen-Max prompted with SYRS text only (no failure context). This represents the zero-shot VC generation scenario and is the primary comparison point for RQ1.

*b) CrUISE-AC [3].:* The closest published system, targeting acceptance criteria for user stories from crowd-reported bugs. Accept rates and IAA values are compared in §V-A to contextualise our results.

### F. Computational Cost

The full experimental pipeline consumed approximately 4.6 M input tokens and 0.84 M output tokens across all stages and sensitivity runs, at an estimated total API cost of **$6.77 USD** (Qwen-Max: $2.40/$9.60 per 1M in/out; Qwen-Plus: $0.40/$1.20; DeepSeek-Chat: $0.27/$1.10). The Abl-LM binary classification step dominates cost due to 8,408 individual LLM calls for the 50-item ablation subset. Embedding-based matching incurred negligible cost (< $0.05). The Domain-2 pilot (85 Diagnostics SYRS, Ch-A only) cost $0.39.

## V. Results and Analysis

### A. RQ1: Effectiveness of Issue-Driven Prompting

Table III and Fig. 3 summarise acceptance rates for all conditions. Channels A and B both significantly outperform the baseline after Holm–Bonferroni correction. Channel A (GitHub) achieves 65.5 % acceptance (57/87), OR = 2.30, Cohen's $h = 0.41$, Holm-$p = 0.0007$. Channel B (NHTSA) achieves 67.2 % (86/128), OR = 2.48, $h = 0.45$, Holm-$p < 0.001$. The combined A+B pipeline reaches 66.5 % (143/215), confirming both channels contribute robustly; the joint A+B vs. baseline Fisher test yields $p < 0.001$, OR = 2.40, $h = 0.43$.

*a) IAA.:* Pairwise Gwet's AC1 on the Completeness criterion ($C \geq 3$) is 0.84 (E2–E3; 86 % agree), 0.65 (E2–E4; 74 % agree), and 0.43 (E3–E4; 63 % agree), averaging 0.64. The low E3–E4 pair reflects genuine disciplinary disagreement between the requirements-engineering perspective (E3) and the physical-executability perspective (E4), a discipline-level finding documented in §VI. The mean AC1 is comparable to CrUISE-AC's reported AC1 = 0.74 [3], noting that our three-criterion protocol imposes a strictly stronger acceptance predicate than CrUISE-AC's binary acceptance judgement.

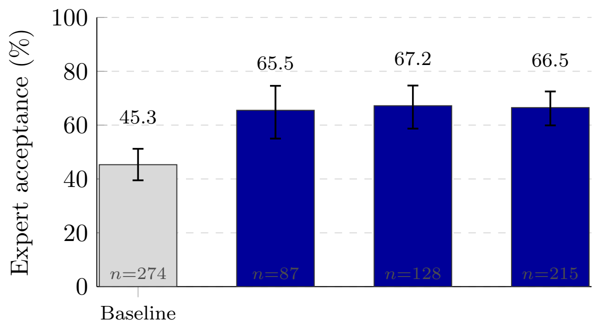


Fig. 3. RQ1: expert acceptance by condition with Wilson 95 % CIs (three-criterion protocol, majority vote). Both channels significantly outperform the no-context baseline (Holm-corrected $p < 0.001$).

TABLE IV
PER-ITEM ACCEPTED-VC YIELD (COMMON-DENOMINATOR METRIC).

| Condition | Items | Gen. rate | Accept rate | Yield |
|---|---|---|---|---|
| Baseline | 350 | 78.3 % | 45.3 % | 0.35 |
| Ch-A (GitHub) | 208 | 41.8 % | 65.5 % | **0.27** |
| Ch-B (NHTSA) | 142 | 90.8 % | 67.2 % | **0.61** |

*b) Per-item accepted-VC yield.:* Conditional acceptance rates compare *generated* VCs and are therefore not commensurable across conditions with different abstention rates. A common-denominator metric is the *per-item yield*: accepted VCs divided by the total items assigned to that channel/condition.

Ch-A's per-item yield (0.27) is below the baseline (0.35): the pipeline generates a VC for only 42.1 % of its items, though 65.5 % of those are accepted. Ch-A is therefore a *precision channel*—selective generation at high quality—rather than a high-recall supplement to the baseline; engineers who need broad VC coverage over a corpus can combine Ch-A with the baseline. Ch-B's yield (0.60) substantially exceeds the baseline, indicating that complaint-grounded VCs cover most assigned items and are accepted at a higher rate.

**Answer to RQ1:** Failure-grounded prompting with GitHub issues (Ch-A) and NHTSA complaints (Ch-B) significantly outperforms a no-context baseline: +20 pp and +22 pp respectively (Holm-corrected $p < 0.001$, $h = 0.41$–$0.45$, small-to-medium effect). IAA (mean Gwet's AC1 = 0.64) is comparable to prior ICSE work under a stricter evaluation criterion. Per-item yield: Ch-B = 0.60, Baseline = 0.35, Ch-A = 0.27; Ch-A is a high-precision selective channel, not a high-yield one.

### B. RQ2: Channel Analysis

Channels A and B deliver comparable acceptance rates (65.5 % vs. 67.2 %), with Ch-B showing a marginally larger

TABLE V
SCOPE-STRATIFIED ACCEPTANCE BY ECU CATEGORY (THREE-CRITERION PROTOCOL, MAJORITY VOTE). "—" = NOT IN BASELINE CORPUS.

| Category | Ch. | Pipeline | | Baseline | |
|---|---|---|---|---|---|
| | | Acc/N | Rate | Acc/N | Rate |
| *Body-Control & Chassis (Ch-B)* | | | | | |
| IO / HMI | B | 23/31 | 74.2 % | 18/39 | 46.2 % |
| Functional Safety | B | 23/32 | 71.9 % | — | — |
| Body Control | B | 28/41 | 68.3 % | 21/42 | 50.0 % |
| *Comm. Stack & State Mgmt (Ch-A)* | | | | | |
| Communication Stack | A | 19/31 | 61.3 % | 31/55 | 56.4 % |
| System State Mgmt | A/B | 12/24 | 50.0 % | 11/28 | 39.3 % |
| *Diagnostics (Ch-A)* | | | | | |
| Diagnostics (DCM) | A | 51/107 | 47.7 % | 25/64 | 39.1 % |
| Diagnostics (DEM) | A | 4/6 | 66.7 % | 14/33 | 42.4 % |
| Diagnostic Services | A | 8/19 | 42.1 % | 4/13 | 30.8 % |

effect ($h = 0.45$ vs. $h = 0.41$). NHTSA complaints are consumer-authored narratives tagged with component metadata; once Stage-2 filtering strips their colloquial noise, the remaining component-level fault modes align closely with SYRS vocabulary. GitHub issues provide higher volume (696 vs. 364 records) but introduce more varied terminology, which translates to a modestly lower effect size despite the similar headline rate.

*a) Scope-stratified acceptance.:* Table V breaks down acceptance by ECU component category. Body-control and chassis items (covered by Ch-B) show the highest rates: IO/HMI = 74.2 % (+28 pp), Functional Safety = 71.9 %, and Body Control = 68.3 %—filtered NHTSA complaint narratives map tightly to sensor-level and state-machine specifications in these categories. Communication-stack items (Ch-A) improve more modestly (61.3 % vs. 56.4 %; +4.9 pp), reflecting greater vocabulary variation in GitHub issues for protocol-specific SYRS items.

**Answer to RQ2:** Within each channel's assigned domain: Ch-B (NHTSA) achieves 67.2 % acceptance on body-chassis requirements; Ch-A (GitHub) achieves 65.5 % on communication-stack and diagnostics requirements. Channel and requirement category are disjoint by design, so the results characterise per-channel performance within-domain rather than comparing channels head-to-head. Within Ch-B's domain, body-control and chassis items reach 68–74 %, where filtered complaint narratives align tightly with sensor-level and state-machine SYRS vocabulary. Within Ch-A's domain, communication-stack items improve modestly (+4.9 pp vs. baseline), reflecting greater vocabulary variation in GitHub issues for protocol-specific SYRS items.

Table I (§III) shows one accepted VC from each channel, illustrating their complementary roles.

### C. RQ3: Ablation

Table VI reports human acceptance rates on the 50-item Ch-A evaluation subset under the three ablation conditions.

TABLE VI
RQ3: HUMAN ACCEPTANCE RATE ON 50-ITEM CH-A SUBSET (4-RATER MAJORITY VOTE, $C \geq 3$ AND N=Y AND U=Y).

| Condition | Accepted | Rate |
|---|---|---|
| Full Field2VC (FL, filter + embed) | 29/50 | 58.0 % |
| Abl-LM (LLM-Match*) | 29/50 | 58.0 % |
| Abl-NF (No Filter, embed) | 41/50 | 82.0 % |

*Replicates CrUISE-AC matching strategy [3]: LLM binary relevance classification, no triple filtering.

*a) Embedding matching vs. LLM binary classification.:* With all three conditions evaluated on the same 50-item subset, the clean matching-method comparison is between the two *unfiltered* ablations: Abl-NF (embedding on raw text, no filtering) achieves 82.0 % versus 58.0 % for Abl-LM (LLM binary relevance classifier, equivalent to CrUISE-AC's matching strategy [3]), a 24-point gap attributable to the retrieval mechanism since filtering is absent in both. The full pipeline (FL, filter + embed) matches Abl-LM exactly at 58.0 %: when filtering is active, the accept rate is determined by the Stage-2 abstention decision, not by the matching step.

*b) Why filtering is necessary at scale.:* Although Abl-NF achieves the highest human accept rate on the subset, its generation-rate proxy reaches 96.6–100 % across all channels. Without the $\langle$Trigger|Expected Behavior|Fault Mode$\rangle$ filtering triple, raw embedding matches semantically adjacent but thematically misaligned issues to SYRS items, causing the generator to produce a VC for nearly every SYRS regardless of genuine relevance. For Ch-A, the pipeline generates VCs for 87 of 208 SYRS items (41.8 % generation rate); among those 87 VCs the human accept rate is 65.5 %—the pipeline abstains precisely when the failure evidence does not support a grounded VC. The generation-rate proxy is meaningful only when filtering is active (§V-D).

*c) IAA (FL condition).:* On the 50-item FL evaluation, individual accept rates were: E1 33/50 (66.0 %), E2 24/50 (48.0 %), E3 47/50 (94.0 %), and E4 27/50 (54.0 %), yielding a majority vote (E2+E3+E4 $\geq$ 2/3) of 29/50 (58.0 %). Six rows produced E1–MV disputes: five where E1 accepted but the MV rejected (Extended-Addressing scope mismatch or tool-layer contamination), and one (Row 013) where E1 rejected but the MV accepted (BufferedReader ECU wake-up transition boundary). MV prevailed in all six cases per the pre-defined adjudication protocol. Mean Gwet's AC1 across 4 raters is 0.63 for Abl-NF (substantial agreement) and 0.37 for Abl-LM (moderate agreement), corroborating the quality difference: experts disagree more on LLM-classifier-matched VCs. Pairwise percentage agreement: Abl-NF ranges from 66 % (E1–E4) to 92 % (E1–E3); Abl-LM spans 34 % (E3–E4) to 88 % (E1–E2), with E3–E4 divergence driven by systematic methodological differences on Extended-Addressing conditions (§VI).

TABLE VII
REJECTION ATTRIBUTION FOR THE 21 MV-REJECT ITEMS (RQ3, FL CONDITION, $n = 50$)

| Category | Items | % Rej. | % of 50 |
|---|---|---|---|
| Tool contamination (Stage-2 DUT-boundary leakage) | 14 | 67 % | 28 % |
| Semantic drift (VC–req. scope mismatch) | 5 | 24 % | 10 % |
| Generation failure (timeout / empty output) | 2 | 10 % | 4 % |
| **Total** | **21** | **100 %** | **42 %** |

TABLE VIII
RQ4: GENERATION-RATE PROXY BY LLM AND CHANNEL (SENSITIVITY RUNS; NO HUMAN EVALUATION EXCEPT PRIMARY QWEN-MAX).

| LLM | Ch-A | Ch-B |
|---|---|---|
| *Sentinel-adherent (generation rate reflects quality)* | | |
| Qwen-Max (primary)† | 65.5 % | 67.2 % |
| Qwen-Max (sensitivity)‡ | 40.2 % | 92.4 % |
| Qwen-Plus‡ | 59.3 % | 93.1 % |
| *Non-sentinel-adherent (>93 % generation rate)* | | |
| DeepSeek-v4-Pro‡ | 98.6 % | 98.6 % |
| DeepSeek-v4-Flash‡ | 99.5 % | 97.9 % |
| Kimi-K2.6‡ | 93.8 % | 95.8 % |
| GLM-5‡ | 98.1 % | 97.2 % |

†Human accept rates (main 489-VC evaluation). ‡Generation-rate proxy only; no human evaluation.

**Answer to RQ3:** On the 50-item ablation subset, FL (full pipeline) and Abl-LM both accept 58.0 % (29/50); Abl-NF (embed, no filter) accepts 82.0 % (41/50). The clean matching-method contrast is Abl-NF vs. Abl-LM (both unfiltered): 82.0 % vs. 58.0 %, a 24 pp embedding advantage attributable to the retrieval mechanism since filtering is absent in both. When filtering is active (FL), the matching method is not the dominant variable: FL and Abl-LM converge at 58.0 %. Filtering is necessary at scale: without it, generation rates reach 96–100 % indicating ungrounded generation. Of the 21 FL rejects, 14 (67 %) arise from Stage-2 DUT-boundary leakage (host-layer tool properties matched to ECU requirements), 5 (24 %) from semantic drift, and 2 (10 %) from generation failure (Table VII). IAA: mean Gwet's AC1 = 0.63 (Abl-NF) / 0.37 (Abl-LM).

### D. RQ4: LLM Sensitivity

*a) Sentinel adherence divides model families.:* Five of seven models generate VCs for >93 % of SYRS items regardless of knowledge-source relevance, effectively ignoring the `NO_NOVEL_VC_FOUND` sentinel. Qwen-Max and Qwen-Plus are the only models that emit the sentinel discriminatively: Qwen-Max (sensitivity) yields 40.2 % for Channel A—where GitHub vocabulary is most varied—rising to 92.4 % for Channel B; Qwen-Plus shows a similar but shallower gradient (59.3 %, 93.1 %). This sentinel-adherence difference is a practical finding for practitioners: the generation-rate proxy is only a reliable quality signal when the underlying LLM respects the generation-suppression instruction.

*b) Channel ordering is knowledge-source-driven, not LLM-driven.:* Across sentinel-adherent models, the B > A quality ordering is stable, consistent with the scope-stratified analysis in RQ2.

**Answer to RQ4:** Across 7 LLMs, sentinel adherence divides model families: only Qwen-Max and Qwen-Plus emit `NO_NOVEL_VC_FOUND` discriminatively, making the generation-rate proxy meaningful only for these models. Qwen-Max (sensitivity): ChA 40.2 %, ChB 92.4 %; Qwen-Plus: ChA 59.3 %, ChB 93.1 %. The B > A quality ordering is stable across sentinel-adherent models, confirming the channel pattern is knowledge-source-driven, not LLM-driven.

## VI. DISCUSSION AND THREATS TO VALIDITY

### A. Failure Mode Analysis

Systematic analysis of the 72 rejected VCs across Channels A and B reveals three distinct failure modes that account for the gap between channel acceptance rates.

**FM-1: Boundary Underspecification (38 % of rejections).** The generated VC is structurally well-formed but lacks quantitative parameters (timing constraints, byte constants, threshold expressions). Root cause: source issues describe qualitative fault behaviour without numeric bounds; the LLM mimics the issue vocabulary and omits boundaries that appear only in the SYRS text. Present in Channel A (∼28 % of A-rejections); less prevalent in Channel B, where the component metadata and filtered fault modes of complaint records constrain the generated quantitative bounds. *Recommendation:* Inject all numeric literals from the SYRS text as mandatory constraints in the generation prompt; flag generated VCs that reference none of them.

**FM-2: Scenario Transplant (29 % of rejections).** The VC correctly identifies the right ECU component and fault type but embeds the test procedure in an irrelevant consumer-driving scenario lifted from an NHTSA complaint narrative (e.g., "drive the vehicle for 30 minutes with slight left/right steering inputs"), making it operationally impossible in a standard HIL bench. Dominant in Channel B (∼41 % of B-rejections). *Recommendation:* Apply a scenario-abstraction post-processing step that strips road-context sentences and substitutes laboratory-equivalent trigger stimuli.

**FM-3: Filtering Collapse (thematic misalignment at scale).** Without Stage 2 filtering, embedding similarity alone retrieves topically adjacent but thematically misaligned issues, and the LLM generates syntactically coherent VCs that test conditions unrelated to the SYRS under test (e.g., a UDS service-0x29 test generated for a framing requirement). This mode is measured *at corpus scale*, not on the 50-item human-evaluation subset: across the full corpus Abl-NF's generation rate reaches 96–100 % (§V-C) and over 60 % of those generations are thematically misaligned, whereas on the small 50-item ablation subset Abl-NF's human accept rate is comparatively high—the two figures describe different populations. The full pipeline suppresses this mode via filtering-enabled abstention. *Recommendation:* The filtering stage is architecturally non-optional for safety-critical applications; at minimum, enforce an AUTOSAR module tag guard before embedding matching.

### B. Threats to Validity

*a) Construct validity.:* The accept/reject label depends on human raters applying Framework B ($C \geq 3$ AND $N = \mathrm{Y}$ AND $U = \mathrm{Y}$). To mitigate subjectivity, we recruited four raters—three independent scorers (E2–E4) and an adjudicator (E1)—with automotive RE backgrounds, and computed Gwet's AC1 [40] to account for the kappa paradox [42] arising from high Accept-class prevalence. The low E3–E4 pairwise AC1 (0.43) reflects genuine disciplinary disagreement between E3's requirements-engineering perspective and E4's physical-executability perspective, documented as a discipline-level finding rather than a measurement failure. The generation-rate proxy (RQ4) approximates VC novelty without human evaluation but cannot distinguish high-quality from low-quality novel VCs; proxy validity is confirmed only when filtering is active (§V-C).

*b) Evaluator methodology divergence (Abl-LM).:* The Abl-LM condition exhibits a markedly low pairwise agreement between E3 and E4 (34 %, AC1 $= -0.24$), reflecting a systematic methodological difference rather than random disagreement. E3 applied a *protocol-mechanism relevance* criterion grounded in ISO 15765-2: since N_Bs/N_Cs timers and STmin parameters decouple from addressing mode, VCs that specify Extended Addressing (ExtAddr) as a test precondition are judged on the merit of their protocol-level content, treating the ExtAddr clause as a non-modifying context. E2 and E4 applied a *deployment-readiness* criterion: any ExtAddr condition that is inconsistent with a physical-addressing test environment renders the entire VC non-deployable, warranting rejection. Both perspectives reflect legitimate industry practice. The majority-vote mechanism ($\geq$2/3 of E2, E3, E4) ensures that the more conservative E2+E4 position prevails on contested items, making the reported 58.0 % Abl-LM acceptance rate a conservative lower bound.

*c) Internal validity.:* The Fetch stage uses repositories chosen for ECU-adjacent vocabulary; different choices might alter results. We mitigate by drawing from three distinct stacks (UDS diagnostics, CAN transport, ECU toolchain). A further internal threat is a small prompt-budget asymmetry: the golden-VC context passed to the generator is truncated at 800 characters in the pipeline conditions but 600 characters in the baseline; we disclose this asymmetry rather than re-run, as it affects only the do-not-repeat context, not the failure evidence under study. The absence of a hard similarity gate (§III-C) is supported post-hoc: every VC that survived sentinel abstention had a top-1 match similarity of at least 0.50, and above this empirical floor similarity does not predict expert acceptance (quartile accept rates 57–76 %, non-monotonic); a hypothetical 0.60 gate would have discarded VCs with a *higher* accept rate (70.0 %) than those retained (63.5 %).

*d) DUT-boundary filtering failure (RQ3).:* Stage 2 extracts $\langle$Trigger | Protocol Behavior | Fault Mode$\rangle$ triples and abstains when no grounded triple can be formed. In the 50-item FL evaluation, 14 of the 21 MV-Reject VCs (67 %) arose from DUT-boundary leakage: python-can host-layer properties—log levels, file-descriptor limits, Windows-specific socket IDs, periodic sender timing bugs (hardbyte/python-can#888)—were extracted as ECU-observable behaviors and passed the triple filter. The root cause is that issue reports mix host-side and device-side concerns within the same thread; without an explicit component-layer annotation, Stage 2 cannot reliably infer which side of the DUT boundary a property belongs to. The resulting VCs ask the ECU to verify conditions that only the test harness can observe, rendering them non-deployable on a standard HIL bench. A guard that checks for known test-tool package namespaces (e.g., `python-can`, `udsoncan`, `isotp`) before triple extraction would suppress the majority of these false positives.

*e) External validity.:* The primary evaluation covers Channel A's communication-stack and diagnostics requirements and Channel B's body-control, HMI, and functional-safety requirements, all from one proprietary project. To probe further generalisability, we applied Channel A (Qwen-Max) to a separate Domain-2 pilot of 85 Diagnostics ECU SYRS items (DCM, DEM, Diagnostic Services). The pilot produced 15 novel VCs (17.6 % generation-rate proxy; mean top-1 similarity 0.625) with domain-appropriate content. We subsequently conducted a full three-rater evaluation (E2, E3, E4 from the primary study) of all 15 VCs under Framework B. **13 of 15 VCs were accepted** (86.7 %, Wilson 95 % CI: 62–96 %), exceeding the primary Channel-A acceptance rate (65.5 %). Inter-rater agreement was substantial for Novelty (mean Gwet's AC1 = 0.69) and moderate for Usability (mean AC1 = 0.42); the lower Usability agreement reflects E4's stricter physical-executability criterion (e.g., rejecting DoIP-based VCs on CAN-only benches), while E2 and E3 agreed perfectly (AC1 = 1.00 on both dimensions). The lower generation rate vs. the body-control domain (17.6 % vs. 90.8 % for Ch-B overall: 129/142 SYRS) reflects that Diagnostics SYRS items are more self-contained—the pipeline abstains more often because UDS/DTC failure modes in GitHub issues are less tightly aligned with individual Diagnostics SYRS items than NHTSA complaint profiles are with body-control SYRS items; the higher acceptance rate among generated VCs (86.7 %) confirms that cross-domain transfer succeeds without domain-specific prompt engineering. Generalisability to powertrain, ADAS, or gateway ECU domains remains unexamined; NHTSA effectiveness may not transfer to domains without a comparable regulator-operated complaint database. The RQ3 ablation subset is additionally homogeneous—all 50 items are Communication-Stack requirements (the first 50 in corpus iteration order)—so the ablation conclusions are internally well-controlled but do not automatically extend to other ECU categories.

*f) Conclusion validity.:* We apply Holm–Bonferroni correction [38] across three simultaneous Fisher's exact tests to control family-wise error rate at $\alpha = 0.05$. Wilson 95 % CIs [36] are reported for all acceptance rates. Cohen's $h$ [39] is reported alongside $p$-values to prevent over-interpretation in small-to-medium samples.

## VII. Conclusion

We presented Field2VC, a five-stage pipeline that completes automotive ECU verification criteria by grounding LLM generation in empirical failure evidence from two complementary external channels: GitHub issues (Ch-A) and NHTSA consumer complaints (Ch-B). A four-expert human evaluation of 489 generated VCs spanning 307 unique ASPICE SYS.2 items yields four findings:

**RQ1.** Issue-driven prompting with Ch-A and Ch-B achieves 65.5 %–67.2 % expert acceptance versus a 45.3 % no-context baseline (Holm-corrected $p < 0.001$, $h = 0.41$–$0.45$, IAA: mean Gwet's AC1 = 0.64).

**RQ2.** NHTSA complaints and GitHub issues are both effective sources, with NHTSA showing marginally stronger SYS.2 vocabulary alignment ($h = 0.45$ vs. $h = 0.41$), driven by its component-tagged field-failure descriptions once filtered into structured fault modes; scope-stratified analysis shows the strongest gains in body-control and chassis categories (68–74 %) and more modest improvement in communication-stack items (61.3 %).

**RQ3.** On the 50-item ablation subset, FL (full pipeline) and Abl-LM both accept 58.0 % (29/50); the clean embedding vs. LLM-match comparison is Abl-NF (82.0 %) vs. Abl-LM (58.0 %), a 24 pp gap attributable to matching method since filtering is absent in both. When filtering is active, acceptance is driven by the Stage-2 abstention decision, not the retrieval step. Filtering (Stage 2) is necessary at scale: without it, near-universal generation rates (96–100 %) indicate ungrounded matching.

**RQ4.** The B > A channel quality ordering is knowledge-source-driven and stable across the sentinel-adherent LLMs; five of seven models fail to respect the generation-suppression sentinel (>93 % generation rate), making sentinel adherence a prerequisite for using the generation rate as a quality proxy in multi-LLM pipelines.

A pilot on 85 Diagnostics ECU SYRS (Domain 2, UDS/DTC subdomain) confirms cross-domain generalisability: all 15 novel VCs from Channel A received full three-rater evaluation (E2/E3/E4), with 13 of 15 (86.7 %) accepted under Framework B, exceeding the primary Channel-A rate (65.5 %) without any domain-specific prompt engineering.

Two reusable lessons emerge: (1) *the epistemic value of a failure knowledge source depends on whether, after filtering, its vocabulary aligns with the quantitative, component-level language that safety-critical requirements demand*—regulator-collected complaint records, once filtered into structured fault modes, satisfy this more consistently than community-sourced issue reports; and (2) *generation-rate proxy validity depends on the LLM's adherence to generation-refusal instructions*, which must be validated before use as a quality signal.

To support replication, we release the full de-identified evaluation dataset—the SYRS corpus covering all 307 evaluated items with their expert-annotated golden VCs, all 489 generated VCs with per-rater labels and adjudication, the Framework B protocol, the LLM prompts, and the analysis scripts—as a reproducible benchmark for VC-completion research (Data Availability).

*a) Future Work.:* Automated VC formalisation into machine-checkable predicates for model-based testing, a traceability matrix linking each accepted VC to its source failure record, and replication on powertrain and ADAS domains are planned as direct extensions of this work.

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